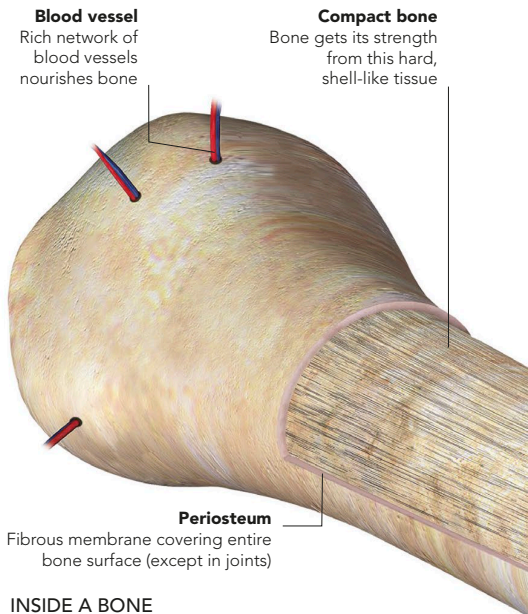


BONE STRUCTURE

BONE IS A TYPE OF CONNECTIVE TISSUE THAT IS AS STRONG AS STEEL BUT AS LIGHT AS ALUMINIUM. IT IS MADE OF SPECIALIZED CELLS AND PROTEIN FIBRES. NEITHER IMMOBILE NOR DEAD, BONE CONSTANTLY BREAKS DOWN AND REBUILDS ITSELF.

STRUCTURE OF A BONE

Along the central shaft of a long bone (such as the femur or humerus) is the medullary canal or marrow cavity. This contains red bone marrow, which produces blood cells; yellow marrow, which is mostly fatty tissue; and blood vessels. Surrounding the marrow cavity is a layer of spongy (cancellous) bone, the honeycomb-like cavities of which also contain marrow. Around this is a layer of compact (cortical) bone, which is hard, dense, and strong. Small canals connect the marrow cavity with the periosteum – a membrane covering the bone surface. Bone tissue is made of specialized cells and protein fibres, woven into a matrix of water, mineral salts, and other substances. Bone cells include osteoblasts, which calcify bone as it forms; osteocytes, which maintain healthy bone structure; and osteoclasts, which absorb bone tissue where it is degenerating or not needed.



INSIDE A BONE

Long bones, for example those in the leg, comprise several different types of bone tissue.

BONE GROWTH

During development in the womb and infancy, most bones develop from structures made of cartilage. Ossification is the process by which this cartilage is converted into bone tissue by the deposition of mineral salts and crystals. Near each end of a long bone is an area called the growth plate, where lengthening and ossification occur. Cartilage cells (see right) multiply here and form columns towards the bone shaft. As the cartilage cells enlarge and die, the space they occupied is filled by new bone cells.

CARTILAGE TO BONE

Initially, ossification in a long bone occurs between the shaft and the head. Later, ossification also takes place inside the head.

